463. NOTE ON A DIFFERENTIAL EQUATION.

[From the Memoirs of the Literary and Philosophical Society of Manchester, vol. II. *(*1865*), pp.* 111–114. *Read February* 18, 1862.*]*

The following investigation was suggested to me by Mr Harley's "Remarks on the Theory of the Transcendental Solution of Algebraic Equations," communicated to the Society at the Meeting of the 4th of February.

Mr Harley's equation

$$y^n - ny + (n-1)\,x = 0$$

may be written

$$y = \frac{n-1}{n}\,x + \frac{1}{n}\,y^n;$$

or putting

$$\frac{n-1}{n}\,x = u, \qquad \frac{1}{n} = a,$$

it becomes

$$y = u + ay^n,$$

which equation may be considered instead of the original equation; and it is to be shown that y, regarded as a function of u, satisfies a certain linear differential equation of the order $n-1$. In fact, expanding y by Lagrange's theorem, we have

$$\begin{aligned} y &= u + au^n + \frac{a^2}{1\cdot 2}\,(u^{2n})' + \frac{a^3}{1\cdot 2\cdot 3}\,(u^{3n})'' + \text{\&c.} \\ &= u + au^n + \frac{a^2}{1\cdot 2}\,2n\,u^{2n-1} + \frac{a^3}{1\cdot 2\cdot 3}\,3n\,(3n-1)\,u^{3n-2} + \text{\&c.}, \end{aligned}$$

the law whereof is obvious, and using the ordinary notation of factorials, viz.

$$[u]^r = u\,(u-1)\,(u-2)\,\ldots\,(u-r+1),$$

we may write

$$y = S_\theta\,\frac{[n\theta]^{\theta-1}}{[\theta]^\theta}\,a^\theta\,u^{n\theta-\theta+1},$$

where θ extends from 0 to ∞.

It is now very easy to show that y satisfies the differential equation

$$\left[u\,\frac{d}{du}\right]^{n-1} y \;=\; na\,\frac{d}{du}\cdot\left[\frac{n}{n-1}\,u\,\frac{d}{du} - \frac{2n-1}{n-1}\right]^{n-1} u^{n-1}\,y.$$

In fact, using on the left-hand side the foregoing value of y, and on the right-hand side the following value of $u^{n-1}\,y$, obtained from that of y by writing $\theta-1$ in the place of θ, viz.

$$u^{n-1}\,y \;=\; S_\theta\,\frac{[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\,a^{\theta-1}\,u^{n\theta-\theta+n},$$

and observing that in general the symbol $u\,\dfrac{d}{du}$, as regards u^m, is $= m$, the equation in question will be satisfied, if only

$$\frac{[n\theta]^{\theta-1}}{[\theta]^\theta}\,\{(n-1)\theta+1\}^{n-1} \;=\; \frac{n\,[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\left[\frac{n}{n-1}\,\{(n-1)\theta+1\} - \frac{2n-1}{n-1}\right]^{n-1},$$

where the right-hand side is

$$= \frac{n\,[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\,\{n\theta-1\}^{n-1};$$

and the equation may be written

$$\frac{n\theta\,[n\theta-1]^{\theta-1}}{\theta\,[\theta-1]^{\theta-1}}\,\{(n-1)\theta+1\}^{n-1} \;=\; \frac{n\,[n\theta-n]^{\theta-2}}{[\theta-1]^{\theta-1}}\,\{n\theta-1\}^{n-1};$$

that is,

$$\{n\theta-1\}^{\theta-1}\,\{(n-1)\theta+1\}^{n-1} \;=\; [n\theta-1]^{n-1}\,\{n\theta-n\}^{\theta-2},$$

which, since each side of the equation is $[n\theta-1]^{\theta+n-2}$, is obviously true.

The foregoing differential equation is developable in the form

$$\left[a_0 + a_1\,u\,\frac{d}{du} + a_2\,u^2\left(\frac{d}{du}\right)^2 + \ldots + a_{n-1}\,u^{n-1}\left(\frac{d}{du}\right)^{n-1}\right]y \;=\; \frac{1}{na}\left(\frac{d}{du}\right)^{n-1} y;$$

but to find the coefficients $a_0, a_1, \ldots, a_{n-1}$, I start from this form, and proceed to substitute in the equation the value of y, which on the left-hand side I use in the original form, and on the right-hand side in the form obtained by writing $\theta+1$ in the place of θ, viz.

$$y \;=\; S_\theta\,\frac{[n(\theta+1)]^\theta}{[\theta+1]^{\theta+1}}\,a^{\theta+1}\,u^{n\theta+n-\theta}.$$

The equation to be satisfied is

$$\frac{[n\theta]^{\theta-1}}{[\theta]^\theta}\Big(a_0 + a_1[(n-1)\theta+1]^1 + a_2[(n-1)\theta+1]^2 + \ldots + a_{n-1}[(n-1)\theta+1]^{n-1}\Big) \;=\; \frac{1}{n}\,\frac{[n(\theta+1)]^\theta}{[\theta+1]^{\theta+1}}\,\{(n-1)\theta+n\}^{n-1},$$

or, what is the same thing,

$$\frac{1}{[\theta]^\theta}\Big(a_0[n\theta]^{\theta-1} + a_1[n\theta]^\theta + a_2[n\theta]^{\theta+1} + \ldots + a_{n-1}[n\theta]^{\theta+n-2}\Big) \;=\; \frac{1}{n}\,\frac{[n(\theta+1)]^{\theta+n-1}}{[\theta+1]^{\theta+1}}.$$

Observing that the right-hand side may be written

$$\frac{1}{n}\,\frac{n(\theta+1)\,[n\theta+n-1]^{\theta+n-2}}{(\theta+1)\,[\theta]^\theta},$$

the equation becomes

$$a_0[n\theta]^{\theta-1} + a_1[n\theta]^\theta + a_2[n\theta]^{\theta+1} + \ldots + a_{n-1}[n\theta]^{\theta+n-2} \;=\; [n\theta+n-1]^{\theta+n-2},$$

or, what is the same thing,

$$a_0 + a_1[(n-1)\theta+1]^1 + a_2[(n-1)\theta+1]^2 + \ldots + a_{n-1}[(n-1)\theta+1]^{n-1} \;=\; [n\theta+n-1]^{n-1};$$

so that $a_0, a_1, \ldots, a_{n-1}$ are the coefficients of the expansion of $[n\theta+n-1]^{n-1}$ (which is a rational and integral function of θ, of the degree $n-1$) in a factorial series, as shown by the left-hand side of the equation.

To determine the actual values, write

$$(n-1)\theta+1=\phi;$$

this gives

$$n\theta+n-1 \;=\; \frac{n\phi+n^2-3n+1}{n-1},$$

and we have therefore

$$\left[\frac{n\phi+n^2-3n+1}{n-1}\right]^{n-1} \;=\; a_0 + a_1[\phi]^1 + a_2[\phi]^2 + \ldots + a_{n-1}[\phi]^{n-1};$$

and thus the general expression is

$$a_r \;=\; \frac{1}{[r]^r}\,\Delta^r\left(\frac{n\phi+n^2-3n+1}{n-1}\right)^{n-1},$$

where Δ denotes the difference in regard to ϕ ($\Delta U_\phi = U_{\phi+1} - U_\phi$), and, after the operation Δ^r is performed, ϕ is to be put equal to zero.

464. NOTE ON PLANA'S LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXIII. *(*1862:1863*), pp.* 211–215.*]*

I have been much surprised to find that there is an error of the order $m^2\gamma^2$, arising from the omission of a factor $(1+\gamma^2)^{-1}$, in the expression for $\dfrac{d^2\delta u}{dv^2}+\delta u$, as given by the equation (II.)' (*Théorie de la Lune*, t. I., p. 267), being the equation made use of in the theory for the determination of δu, the perturbation of the reciprocal of the radius vector. This error may probably be the cause of some of the discrepancies in the terms of the fourth and higher orders, between Plana's results and those of Pontécoulant and Delaunay.

Plana's equation (6), t. I., p. 260, is

$$\begin{aligned}\frac{d^2\delta u}{dv^2}+\delta u = aR''&\\ &+f(e,\gamma)\,Q'e\cos\Big(cv-\int\varpi\,dv\Big)\\ &-\left\{f(e,\gamma)(1+\gamma^2)(1+s_1^2)^{-\frac{3}{2}}-\frac{a(1+s^2)^{-\frac{3}{2}}}{a_1\,\psi(e,\gamma)}\right\}\\ &+f(e,\gamma)\,P\gamma^2\,(1+s_1^2)^{-\frac{3}{2}}\,\Theta,\end{aligned}$$

if for shortness

$$\Theta \;=\; \tfrac{1}{2}\gamma^2-\big(1+\tfrac{1}{2}\gamma^2\big)\cos\Big(2gv-2\int\theta\,dv\Big)+\tfrac{1}{2}\gamma^2\cos\Big(4gv-4\int\theta\,dv\Big).$$

R'' (p. 256) should be

$$R''\;=\;\frac{1}{aa_1\,\psi(e,\gamma)}\cdot\frac{\Omega_1\,\dfrac{du}{dv}\,\dfrac{d\Omega}{u\,dv}-\pi\,(1+s^2)^{-\frac{3}{2}}\,2\textstyle\int U\,dv}{1+2\textstyle\int U\,dv},$$

but, by an error which is implicitly corrected, the σ which multiplies $(1+s^2)^{-\frac{3}{2}}\,2\int U\,dv$ is omitted. Hence the equation (6) becomes

$$\begin{aligned}\Big(1+2\int U\,dv\Big)\left(\frac{d^2\delta u}{dv^2}+\delta u\right) &= \frac{a}{aa_1\,\psi(e,\gamma)}\left\{\Omega_1-\frac{du}{dv}\,\frac{d\Omega}{u\,dv}-\sigma\,(1+s^2)^{-\frac{3}{2}}\,2\int U\,dv\right\}\\ &+\Big(1+2\int U\,dv\Big)\,f(e,\gamma)\,Q'e\cos\Big(cv-\int\varpi\,dv\Big)\\ &-\Big(1+2\int U\,dv\Big)\left\{f(e,\gamma)(1+\gamma^2)(1+s_1^2)^{-\frac{3}{2}}-\frac{a(1+s^2)^{-\frac{3}{2}}}{a_1\,\psi(e,\gamma)}\right\}\\ &+\Big(1+2\int U\,dv\Big)\,f(e,\gamma)\,P\gamma^2(1+s_1^2)^{-\frac{3}{2}}\,\Theta,\end{aligned}$$

in which equation

$$\Omega_1=\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv},\quad \text{pp. 26, 245,}\qquad \frac{d\Omega}{u^t\,dv}=\frac{\sigma\sigma_t}{\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}}\,U,\quad \text{p. 265,}$$

$$\psi(e,\gamma)=\lambda^{-\frac{1}{2}}(1+\gamma^2)^{-\frac{1}{2}},\qquad f(e,\gamma)=\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}},\quad \text{p. 261.}$$

But retaining for greater convenience the function $f(e,\gamma)$ in two of the terms, we have

$$\begin{aligned}\Big(1+2\int U\,dv\Big)\left(\frac{d^2\delta u}{dv^2}+\delta u\right) = \frac{a}{aa_1\,\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}}\Big[&\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv}\\ &-\frac{\sigma\sigma_t}{\lambda^{\frac{1}{2}}(1+\gamma^2)^{\frac{1}{2}}}\,U\,\frac{du}{dv}-\pi(1+s^2)^{-\frac{3}{2}}\,2\int U\,dv\Big]\end{aligned}$$

$$
\begin{aligned}
&+\left(1+2\int U\,dv\right) f(e,\gamma)\, Q'e\cos\left(cv-\int\varpi\,dv\right)\\
&-\left(1+2\int U\,dv\right)\lambda^{\frac12}(1+\gamma^2)^{\frac32}\left\{(1+s_1^2)^{-\frac32}-\frac{a}{a_1}\,(1+s^2)^{-\frac32}\right\}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\, P\gamma^2(1+s_1^2)^{-\frac32}\,\Theta,
\end{aligned}
$$

which is

$$
\begin{aligned}
&=\frac{a}{aa_1}\,\lambda^{\frac12}(1+\gamma^2)^{\frac12}\left(\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv}\right)\\
&-a\,U\,\frac{du}{dv}\\
&-\frac{a}{a_1}\,\lambda^{\frac12}(1+\gamma^2)^{\frac12}\,(1+s^2)^{-\frac32}\,2\int U\,dv \quad \text{(destroyed by term } \textit{infra}\text{)}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\, Q'e\cos\left(cv-\int\varpi\,dv\right)\\
&-\lambda^{\frac12}(1+\gamma^2)^{\frac32}\left\{(1+s_1^2)^{-\frac32}-\frac{a}{a_1}(1+s^2)^{-\frac32}\right\}\\
&-\lambda^{\frac12}(1+\gamma^2)^{\frac32}\,(1+s_1^2)^{-\frac32}\,2\int U\,dv\\
&+\frac{a}{a_1}\,\lambda^{\frac12}(1+\gamma^2)^{\frac12}\,(1+s^2)^{-\frac32}\,2\int U\,dv \quad \text{(destroyed by term } \textit{supra}\text{)}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\, P\gamma^2(1+s_1^2)^{-\frac32}\,\Theta;
\end{aligned}
$$

or, putting $u=\dfrac{1}{a}\,(\nu+\delta u)$, this becomes

$$
\begin{aligned}
\left(1+2\int U\,dv\right)\left(\frac{d^2\delta u}{dv^2}+\delta u\right)&=\frac{a}{aa_1}\,\lambda^{\frac12}(1+\gamma^2)^{\frac12}\left(\frac{d\Omega}{du}+\frac{s}{u}\,\frac{d\Omega}{dv}\right)\\
&-U\left(\frac{du}{dv}+\frac{d\delta u}{dv}\right)\\
&-\lambda^{\frac12}(1+\gamma^2)^{\frac32}\,(1+s_1^2)^{-\frac32}\,2\int U\,dv\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\, Q'e\cos\left(cv-\int\varpi\,dv\right)\\
&-\lambda^{\frac12}(1+\gamma^2)^{\frac32}\left\{(1+s_1^2)^{-\frac32}-\frac{a}{a_1}(1+s^2)^{-\frac32}\right\}\\
&+\left(1+2\int U\,dv\right) f(e,\gamma)\, P\gamma^2(1+s_1^2)^{-\frac32}\,\Theta,
\end{aligned}
$$

agreeing with the Formula II. p. 265, except that in Plana's last term, instead of the factor $f(e,\gamma)$ $(=\lambda^{\frac12}(1+\gamma^2)^{\frac12})$, we have the factor $\lambda^{\frac12}(1+\gamma^2)^{\frac32}$. That is, the last term, as given by Plana, should be divided by $1+\gamma^2$. And this error is introduced from the formula (II.) into the formula (II.)', p. 267, viz. the incorrect factor $\lambda^{\frac12}(1+\gamma^2)^{\frac12}$ is there replaced by its value q; whereas, the true value being $\lambda^{\frac12}(1+\gamma^2)^{\frac32}$, the factor in (II.)' should be $\dfrac{q}{1+\gamma^2}$.

The corrected formula (II.)' is

$$
\begin{aligned}
-\frac{d^2\delta u}{dv^2}-\delta u=-Q'\,\frac{qe}{1+\gamma^2}\,&\cos\left(cv-\int\varpi\,dv\right)\\
&+q\left\{(1+s_1^2)^{-\frac32}-\frac{a}{a_1}(1+s^2)^{-\frac32}\right\}\\
&+\mu^2(R_2+R_4)-\mu^2\left(\frac{du}{dv}+\frac{d\delta u}{dv}\right)R_1
\end{aligned}
$$

$$- 2\mu^2\left[\frac{d^2\delta u}{dv^2} + \delta u + q\,(1+s_1^2)^{-\frac{3}{2}} - \frac{Q'qe}{1+\gamma^2}\cos\left(cv - \int\varpi\,dv\right)\right]\int R_1\,dv$$
$$- Pq\gamma^2\,(1+\gamma^2)^{-1}\,(1+s_1^2)^{-\frac{3}{2}}\,(1-2\mu^2\int R_1\,dv)\times$$
$$\left\{\tfrac{1}{2}\gamma^2 - (1+\tfrac{1}{2}\gamma^2)\cos(2gv - 2\int\theta dv) + \tfrac{1}{2}\gamma^2\cos(4gv - 4\int\theta dv)\right\}.$$

Observing that P is of the order m^2, and that q is approximately equal to unity, the error in $\dfrac{d^2\delta u}{dv^2} + \delta u$ is of the order $m^2\gamma^2$, as noticed above. It may be right to mention that I obtained the correction in the first instance by starting from the fundamental equations, and not as here from the intermediate equation (6), so that there is not in that equation any error afterwards implicitly corrected in the transformation to (II.)'.

465. NOTE ON THE LUNAR THEORY.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXV. *(*1864–1865*), pp.* 182–189.*]*

I attend, in the expressions for the lunar coordinates, only to the coefficients independent of m. Plana's values, taken to the fourth order only, are as follows; for greater simplicity I write $a = 1$; and, instead of $nt +$ constant, $cnt +$ constant, $gnt +$ constant, I write g, l, c, respectively; viz., l is the mean longitude, c the mean anomaly, g the mean distance from node: this being so, then r, v, y, denoting the radius vector, longitude and latitude respectively, we have

$$\frac{1}{r}\,(\text{Plana}) = \begin{array}{rll} & & \cos\ c \\ +e - \frac{1}{8}e^3 - \frac{1}{2}\gamma^2 e & ,, & c \\ +e^2 - \frac{1}{3}e^4 - \frac{1}{2}\gamma^2 e^2 & ,, & 2c \\ +\frac{3}{8}e^3 & ,, & 3c \\ +\frac{1}{3}e^4 & ,, & 4c \\ -\frac{1}{2}\gamma^2 & ,, & 2g \\ -\frac{1}{8}\gamma^2 e & ,, & c-2g \end{array}$$

(but I omit Plana's term $+\frac{3}{8}\gamma^2 e\,\cos(2c+2g)$ which should be $= 0$).

$$v\,(\text{Plana}) = l + \begin{array}{rll} & & \sin\ c \\ +2e - \frac{1}{4}e^3 - \frac{1}{2}\gamma^2 e & ,, & c \\ +\frac{5}{4}e^2 - \frac{11}{24}e^4 - \frac{3}{4}\gamma^2 e^2 & ,, & 2c \\ +\frac{13}{12}e^3 & ,, & 3c \\ +\frac{103}{96}e^4 & ,, & 4c \\ -\frac{1}{4}\gamma^2 - \frac{1}{16}\gamma^2 e^2 + \frac{1}{4}\gamma^4 & ,, & 2g \\ +\frac{1}{4}\gamma^2 e & ,, & c-2g \\ -\frac{1}{8}\gamma^2 e & ,, & c+2g \\ -\frac{1}{16}\gamma^2 e^2 & ,, & 2c-2g \\ +\frac{11}{48}\gamma^2 e^2 & ,, & 2c+2g \\ +\frac{1}{32}\gamma^4 & ,, & 4g \end{array}$$

$$y\,(\text{Plana}) = \begin{array}{rll} & & \sin\ g \\ \gamma - \gamma e^2 - \frac{3}{8}\gamma^4 & ,, & g \\ +\gamma e - \frac{1}{8}\gamma e^3 & ,, & c-g \\ +\gamma e - \frac{7}{8}\gamma e^3 - \frac{1}{2}\gamma^3 e & ,, & c+g \\ +\frac{17}{8}\gamma e^2 & ,, & 2c-g \\ +\frac{5}{8}\gamma e^2 & ,, & 2c+g \\ +\frac{17}{12}\gamma e^3 & ,, & 3c-g \\ +\frac{5}{12}\gamma e^3 & ,, & 3c+g \\ -\frac{1}{8}\gamma^3 & ,, & 3g \\ +\frac{1}{16}\gamma^3 e & ,, & c-3g \\ -\frac{1}{16}\gamma^3 e & ,, & c+3g. \end{array}$$

To compare these with the elliptic values, it is necessary to write $e\,(1+\frac{1}{2}\gamma^2)$ in place of e. Making this change, or say reducing Plana's (e,γ) to the elliptic (e,γ), I write down in a first column the transformed coefficients, and in a second column the elliptic coefficients, as follows:

Plana, with Elliptic e,γ	Elliptic	
$\frac{1}{r}=$	$\frac{1}{r}=$	
1	1	$\cos\ c$
$+e-\frac{1}{8}e^3$	$+e-\frac{3}{8}e^3$	„ c
$+e^2-\frac{1}{3}e^4$	$+e^2-\frac{1}{3}e^4$	„ $2c$
$+\frac{3}{8}e^3$	$+\frac{3}{8}e^3$	„ $3c$
$+\frac{1}{3}e^4$	$+\frac{1}{3}e^4$	„ $4c$
$-\frac{1}{2}\gamma^2$	0	„ $2g$
$-\frac{1}{8}\gamma^2 e$	0	„ $c-2g$.

Plana, with Elliptic e,γ	Elliptic	
$v=$	$v=$	
l	l	$\sin\ c$
$+2e-\frac{1}{4}e^3$	$+2e-\frac{1}{4}e^3$	„ c
$+\frac{5}{4}e^2-\frac{11}{24}e^4-\frac{1}{16}\gamma^2e^2$	$+\frac{5}{4}e^2-\frac{11}{24}e^4$	„ $2c$
$+\frac{13}{12}e^3$	$+\frac{13}{12}e^3$	„ $3c$
$+\frac{103}{96}e^4$	$+\frac{103}{96}e^4$	„ $4c$
$-\frac{1}{4}\gamma^2-\frac{1}{16}\gamma^2e^2+\frac{1}{4}\gamma^4$	$-\frac{1}{4}\gamma^2$	„ $2g$
$+\frac{1}{4}\gamma^2 e$	$-\frac{1}{4}\gamma^2 e$	„ $c-2g$
$-\frac{1}{8}\gamma^2 e$	$+\frac{1}{16}\gamma^2 e$?	„ $c+2g$
$-\frac{1}{16}\gamma^2e^2$		„ $2c-2g$
$+\frac{11}{48}\gamma^2e^2$		„ $2c+2g$
$+\frac{1}{32}\gamma^4$		„ $4g$.

Plana, with Elliptic e,γ	Elliptic	
$y=$	$y=$	
$\gamma-\gamma e^2-\frac{3}{8}\gamma^3$	$+\gamma e-\gamma e^2-\frac{1}{2}\gamma^3$	$\sin\ g$
$+\gamma e-\frac{3}{8}\gamma e^3-\frac{3}{8}\gamma^3 e$	$+\gamma e-\frac{17}{8}\gamma e^3-\frac{1}{2}\gamma^3 e$	„ $c+g$
$+\gamma e-\frac{7}{8}\gamma e^3+\frac{1}{2}\gamma^3 e$	$+\gamma e-\frac{17}{8}\gamma^3 e$	„ $c-g$
$+\frac{2}{3}\gamma e^2$		
$+\frac{1}{8}\gamma e^2$	$+\frac{1}{8}\gamma e^2$	„ $2c-g$
$+\frac{1}{16}\gamma e^2$		
$+\frac{17}{12}\gamma e^3$	$+\frac{17}{12}\gamma e^3$	„ $3c-g$
$+\frac{5}{12}\gamma e^3$	$+\frac{5}{12}\gamma e^3$	„ $3c+g$
$-\frac{1}{8}\gamma^3$	$-\frac{1}{8}\gamma^3$	„ $3g$
$+\frac{1}{16}\gamma^3 e$	$-\frac{1}{16}\gamma^3 e$	„ $c-3g$
$-\frac{1}{16}\gamma^3 e$	$+\frac{1}{16}\gamma^3 e$	„ $c+3g$,

where, for greater clearness, I remark that the values called "elliptic" of e, γ, c, g, refer to an ellipse, such that the longitude of the node, and the longitude (in orbit) of the pericentre, vary uniformly with the time, viz., we have mean distance $=1$, excentricity $=e$, tangent of inclination $=\gamma$, mean longitude $=l$, mean anomaly $=c$, distance from node $=g$.

We have therefore

$$
\begin{array}{rlll}
\delta\frac{1}{r} = & -\tfrac{1}{2}\gamma^2e^2 & \cos & 2g \\
 & -\tfrac{1}{8}\gamma^2 e & ,, & c-2g, \\
\delta v = & -\tfrac{1}{16}\gamma^2e^2 & \sin & 2c \\
 & -\tfrac{11}{48}\gamma^2e^2 & ,, & 2g \\
 & +\tfrac{1}{4}\gamma^2 e & ,, & c-2g \\
 & -\tfrac{1}{16}\gamma^2e^2 & ,, & 2c-2g, \\
\delta y = & -\tfrac{3}{8}\gamma^3+\tfrac{1}{8}\gamma e^2 & \sin & c-g \\
 & +\tfrac{2}{3}\gamma e^2 & ,, & 2c-g \\
 & +\tfrac{2}{3}\gamma e^3 & ,, & 3c-g \\
 & +\tfrac{1}{8}\gamma^3 e & ,, & c-3g,
\end{array}
$$

viz., these are the increments to be added to the elliptic values of $\frac{1}{r}$, v, y, respectively, in order to obtain the disturbed values of $\frac{1}{r}$, v, y, attending only to the coefficients independent of m; *they represent, in fact, the Lunar inequalities which rise two orders by integration.*

The elliptic values of $\frac{1}{r}$ and y are functions, and that of v, is equal to $l+$ a function, of e, γ, c, g, and the foregoing disturbed values may be obtained by affecting each of the quantities e, γ, c, g, and l, with an inequality depending on the argument $2c-2g$, viz., these inequalities are

$$
\begin{aligned}
\delta e &= -\tfrac{3}{8}\gamma^2 e\cos 2c-2g \\
\delta c &= \tfrac{1}{8}\gamma^2\sin 2c-2g \\
\delta\gamma &= -\tfrac{3}{8}\gamma e^2\cos 2c-2g \\
\delta g &= \tfrac{3}{8}e^2\sin 2c-2g \\
\delta l &= -\tfrac{1}{16}\gamma^2e^2\sin 2c-2g.
\end{aligned}
$$

The verification may be effected without difficulty; thus, for instance, starting from the elliptic value of $\frac{1}{r}$, we have to the fourth order

$$
\begin{aligned}
\delta\frac{1}{r} &= \delta\begin{pmatrix} e\cos c \\ +e^2\cos 2c\end{pmatrix}\begin{pmatrix} -e\sin c \\ -2e^2\sin 2c\end{pmatrix}\delta c' + \begin{pmatrix}\cos c \\ +2e\cos 2c\end{pmatrix}\delta e \\
&= -\tfrac{1}{8}\gamma e\,(-\sin c\,\sin 2c-2g\,\cos c\,\cos 2c-2g) \\
&\quad +\tfrac{3}{8}\gamma^2\,(-\sin 2c\,\sin 2c-2g\; - \;\cos 2c\,\cos 2c-2g) \\
&= -\tfrac{1}{2}\gamma^2 e\,\cos c-g \\
&\quad -\tfrac{3}{8}\gamma^2\,\cos 2g,
\end{aligned}
$$

which is right; and the verification of the values of δv, δy, may be effected in a similar manner.

I have, in order to fix the ideas, preferred to give in the first instance the foregoing *à posteriori* proof; but I now inquire generally as to the form of the values of $\frac{1}{r}$, v, y, or say of r, v, y, taking account only of coefficients independent of m; and I proceed to show that these may be obtained from the elliptic values expressed as above in terms of l, e, γ, c, g, by affecting l, e, γ, c, g, each with an inequality depending on the multiple sines or cosines of $c-g$.

Writing for greater simplicity $n=1$, we have $l=t+L$, $c=ct+C$, $g=gt+G$, where $c=1-\frac{3}{4}m^2+\&c.$, $g=1+\frac{3}{4}m^2+\&c.$; viz., c, g, are constants which differ from unity by terms involving m^2.

The required values of r, v, y, satisfy the undisturbed equations of motion, if after the differentiations we write in the coefficients (which coefficients are functions of m through c, g) $m=0$; that is, if we write in the coefficients $c=1$, $g=1$. In fact, the required values of r, v, y, are what the complete values become, upon writing in the coefficients of the complete values $m=0$; that is, the required values of r, v, y, differ from the complete values by terms the coefficients whereof contain m as a factor; and the disturbed equations differ from the undisturbed equations in that they contain the differential coefficients of the disturbing function; that is, terms the coefficients whereof have the factor m^2. Imagine the complete values of r, v, y, substituted in the disturbed equations of motion; the resulting equations are satisfied identically; and, therefore, whatever be the value of m; that is, they are satisfied if in these equations respectively we write $m = 0$: it requires a little consideration to see that this is so, if in *the coefficients only* we write $m = 0$; but recollecting that c, g, stand for functions $ct + C$, $gt + G$, so that, for example, $c - g, = (c - g)t + C - G$, upon writing therein $m = 0$, becomes equal, not to zero, but to the constant value $C - G$, the identity subsists in regard to the coefficient of the sine or cosine of each separate argument $ac \pm \beta g$, and, consequently, it subsists notwithstanding that in the arguments c and g, instead of being each put $= 1$, are left indeterminate. And granting this (viz. that the equations are satisfied if in the coefficients only we write $m = 0$), then it is clear that, as above stated, the required values of r, v, y, satisfy the undisturbed equations of motion, if after the differentiations we write in the coefficients $c = 1$, $g = 1$.

The required values of r, v, y, are of the form $r = \phi(c, g)$, $y = \psi(c, g)$, $v = l + \chi(c, g)$, but writing $w = v + c - l, = c + \chi(c, g)$, the last mentioned property will equally subsist in regard to the functions r, w, y: in fact, v enters into the differential equations only through its differential coefficient $\frac{dv}{dt}$, and the differential coefficients of v and w, that is, of $l + \chi(c, g)$ and $c + \chi(c, g)$, differ only by the quantity $c - 1$, which becomes $= 0$, in virtue of the assumed relations $c = 1$, $g = 1$.

Hence the undisturbed equations are satisfied by the values $r = \phi(c, g)$, $y = \psi(c, g)$, $w = c + \chi(c, g)$, when after the differentiations we write in the coefficients $c = 1$, $g = 1$; the foregoing values contain t through the quantities c, g, only; and we have, therefore,

$$\frac{d}{dt} = \frac{d}{dc} + \frac{d}{dg}.$$

Hence, writing in the coefficients $c = 1$, $g = 1$, we have $\frac{d}{dt} = \frac{d}{dc} + \frac{d}{dg}$; that is, the values $r = \phi(c, g)$, $y = \psi(c, g)$, $w = \chi(c, g)$, regarding r, v, y, as functions of c, g, satisfy the partial differential equations obtained from the undisturbed equations of motion by writing therein $\frac{d}{dc} + \frac{d}{dg}$ in place of $\frac{d}{dt}$. Hence also, considering r, w, y, as functions of c and $c - g$, then observing that $\left(\frac{d}{dc} + \frac{d}{dg}\right)(c - g) = 0$, the values of r, v, y, satisfy the partial differential equations obtained by writing $\frac{d}{dc}$ in place of $\frac{d}{dt}$; and inasmuch as these partial differential equations do not contain $\frac{d}{dg}$, they are to be integrated as ordinary differential equations in regard to c as the independent variable, the constants of integration being replaced by arbitrary functions of $c - g$.

Consider the pure elliptic values of r, v, y, in an elliptic orbit with the following elements: A, the mean distance; N, the mean motion ($N^2A^3 = 1$ and therefore $A = N^{-\frac{2}{3}}$); E, the excentricity; $Nt + D$, the mean anomaly; $Nt + H$, the mean distance from node; $Nt + K$, the mean longitude; then writing c in place of t, we have

$$\begin{aligned} r &= N^{-\frac{2}{3}} \operatorname{elqr}(E,\, Nc + D), \\ v\,(= l - c + w) &= l - c + Nc + K + P(E,\, \Gamma,\, Nc + D,\, Nc + H), \\ y &= Q(E,\, \Gamma,\, Nc + D,\, Nc + H), \end{aligned}$$

where N, E, Γ, D, H, K, are arbitrary functions of $c - g$: P and Q denote given functional expressions. But, in order that r, v, y, considered as functions of c and g, may be of the proper form, it is necessary as regards N to write simply $N = 1$; we have then

$$\begin{aligned} r &= \operatorname{elqr}(E,\, c + D), \\ v &= l + K + P(E,\, \Gamma,\, c + D,\, c + H), \\ y &= Q(E,\, \Gamma,\, c + D,\, c + H), \end{aligned}$$

where E, Γ, D, H, K, are arbitrary functions of $c - g$; or, what is the same thing, writing for these quantities respectively $e + \delta e$, $\gamma + \delta\gamma$, $c + \delta c$, $g + \delta g$, $l + \delta l$, where δe, $\delta\gamma$, δc, δg, δl are arbitrary functions of $c - g$, we have

$$\begin{aligned} r &= \operatorname{elqr}(e + \delta e,\, c + \delta c), \\ v &= l + \delta l + P(e + \delta e,\, \gamma + \delta\gamma,\, c + \delta c,\, g + \delta g), \end{aligned}$$

$$y = Q(e + \delta e,\ \gamma + \delta\gamma,\ c + \delta c,\ g + \delta g),$$

that is, the values of r, v, y, are obtained from the elliptic values

$$\begin{aligned} r &= \mathrm{elqr}\,(e,\ c), \\ v &= l + P(e,\ \gamma,\ c,\ g), \\ y &= Q(e,\ \gamma,\ c,\ g), \end{aligned}$$

by affecting each of the quantities $e,\ \gamma,\ c,\ g,\ l$, with an inequality which is a function of $c - g$.

466. Second Note on the Lunar Theory.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXV. (1864–1865), pp. 203–207.]

The elliptic values of

r, the radius vector, v, the longitude, y, the latitude,

are functions of

a, the mean distance,
e, the excentricity,
γ, the tangent of the inclination,
l, the mean longitude,
c, the mean anomaly,
g, the mean distance from node;

see my Note in the last Monthly Notice, p. 182, [465], where, for the present purpose, $\dfrac{a}{r}$ should be written instead of $\dfrac{1}{r}$; and it is there shown that the disturbed values, attending only to the coefficients independent of m, are obtained by affecting $a,\ e,\ \gamma,\ c,\ g,\ l$, with the inequalities

$$\begin{aligned} \delta a &= 0 \\ \delta e &= -\tfrac{3}{8}\gamma^2 e \cos 2c - 2g \\ \delta\gamma &= +\tfrac{3}{8}\gamma e^2 \quad ,, \quad 2c - 2g \\ \delta c &= +\tfrac{1}{8}\gamma^2 \sin 2c - 2g \\ \delta g &= +\tfrac{3}{8}e^2 \quad ,, \quad 2c - 2g \\ \delta l &= -\tfrac{1}{16}\gamma^2 e^2\ ,, \quad 2c - 2g, \end{aligned}$$

or, what is the same thing, adding to the elliptic values the inequalities

$$\begin{aligned} \delta\frac{a}{r} &= -\tfrac{1}{2}\gamma^2 \cos 2g \\ &\quad - \tfrac{1}{8}\gamma^2 e \quad ,, \quad c - 2g, \\ \delta v &= -\tfrac{1}{16}\gamma^2 e^2 \sin 2c \\ &\quad - \tfrac{11}{48}\gamma^2 e^2 \quad ,, \quad 2g \\ &\quad + \tfrac{1}{4}\gamma^2 e \quad ,, \quad c - 2g \\ &\quad - \tfrac{1}{16}\gamma^2 e^2 \quad ,, \quad 2c - 2g, \\ \delta y &= -\tfrac{3}{8}\gamma^3 + \tfrac{1}{8}\gamma e^2 \sin c - g \\ &\quad + \tfrac{2}{3}\gamma e^2 \quad ,, \quad 2c - g \\ &\quad + \tfrac{2}{3}\gamma e^3 \quad ,, \quad 3c - g \\ &\quad + \tfrac{1}{8}\gamma^3 e \quad ,, \quad c - 3g. \end{aligned}$$

I propose to show how these results may be obtained by the method of the variation of the elements. For this purpose, treating a, e, γ, c, g, l, as elements, the proper formulae are obtained very readily from those given in my "Memoir on the Problem of Disturbed Elliptic Motion," Mem. R. Ast. Soc., vol. XXVII. (1859), pp. 1–29, [212]; viz., writing c in place of g, the formulae, p. 25, give the variations of a, e, c, η, θ, ϕ; we have then

$$g = c + \mathfrak{C}, \qquad l = c + \mathfrak{C} + \theta,$$

and therefore

$$\gamma = \tan\phi,$$

$$\begin{aligned} dg &= dc + d\mathfrak{C}, \\ dl &= dc + d\mathfrak{C} + d\theta, \\ d\gamma &= (1+\gamma^2)\, d\phi, \end{aligned}$$

which give for the transformation of the differential coefficients of Ω,

$$\begin{aligned} \frac{d\Omega}{dc} &= \frac{d\Omega}{dc} + \frac{d\Omega}{dg} + \frac{d\Omega}{dl}, \\ \frac{d\Omega}{d\mathfrak{C}} &= \frac{d\Omega}{dg} + \frac{d\Omega}{dl}, \\ \frac{d\Omega}{d\theta} &= \frac{d\Omega}{dl}, \\ \frac{d\Omega}{d\phi} &= (1+\gamma^2)\,\frac{d\Omega}{d\gamma}, \end{aligned}$$

and the formulae finally become

$$\begin{aligned} \frac{da}{dt} &= \frac{2}{na}\frac{d\Omega}{dc} + \frac{2}{na}\frac{d\Omega}{dg} + \frac{2}{na}\frac{d\Omega}{dl}, \\ \frac{de}{dt} &= \frac{1-e^2}{na^2e}\frac{d\Omega}{dc} + \frac{1-e^2-\sqrt{1-e^2}}{na^2e}\frac{d\Omega}{dg} + \frac{1-e^2-\sqrt{1-e^2}}{na^2e}\frac{d\Omega}{dl}, \\ \frac{d\gamma}{dt} &= \frac{1+\gamma^2}{na^2\sqrt{1-e^2}\,\gamma}\frac{d\Omega}{dg} + \frac{(1+\gamma^2)(1-\sqrt{1+\gamma^2})}{na^2\sqrt{1-e^2}\,\gamma}\frac{d\Omega}{dl}, \\ \frac{dc}{dt} &= -\frac{2}{na}\frac{d\Omega}{da} - \frac{1-e^2}{na^2e}\frac{d\Omega}{de}, \\ \frac{dg}{dt} &= -\frac{2}{na}\frac{d\Omega}{da} - \frac{1-e^2-\sqrt{1-e^2}}{na^2e}\frac{d\Omega}{de} - \frac{1+\gamma^2}{na^2\sqrt{1-e^2}\,\gamma}\frac{d\Omega}{d\gamma}, \\ \frac{dl}{dt} &= -\frac{2}{na}\frac{d\Omega}{da} - \frac{1-e^2-\sqrt{1+e^2}}{na^2e}\frac{d\Omega}{de} - \frac{(1+\gamma^2)(1-\sqrt{1+\gamma^2})}{na^2\sqrt{1-e^2}\,\gamma}\frac{d\Omega}{d\gamma}. \end{aligned}$$

The disturbing function contains the term

$$m^2a^2a''\left(+\tfrac{3}{8}e^2\gamma^2\right)\cos 2c-2g.$$

If after the differentiations we write for greater simplicity $a=1$, $n=1$, we have

$$\begin{aligned} \frac{d\Omega}{da} &= +\tfrac{15}{4}\,m^2e^2\gamma^2\cos 2c-2g, \\ \frac{d\Omega}{de} &= +\tfrac{3}{4}\,m^2e\gamma^2\ ,,\ 2c-2g, \\ \frac{d\Omega}{d\gamma} &= +\tfrac{3}{4}\,m^2e^2\gamma\ ,,\ 2c-2g, \\ \frac{d\Omega}{dc} &= -\tfrac{3}{4}\,m^2e^2\gamma^2\sin 2c-2g, \\ \frac{d\Omega}{dg} &= -\tfrac{3}{4}\,m^2e^2\gamma^2\ ,,\ 2c-2g, \end{aligned}$$

$$\frac{d\Omega}{dl} = 0,$$

and the formulae for the variations give

$$\begin{aligned}
\frac{da}{dt} &= \frac{2}{na}\left(\frac{d\Omega}{dc} + \frac{d\Omega}{dg}\right) = 0,\\
\frac{de}{dt} &= -\frac{1}{e}\frac{d\Omega}{dc} = -\tfrac{3}{8}\,m^2e\gamma^2 \sin 2c-2g,\\
\frac{d\gamma}{dt} &= -\frac{1}{\gamma}\frac{d\Omega}{dg} = -\tfrac{3}{8}\,m^2\gamma e^2 \text{ ,, } 2c-2g,\\
\frac{dc}{dt} &= -\frac{1}{e}\frac{d\Omega}{de} = -\tfrac{3}{8}\,m^2\gamma^2 \cos 2c-2g,\\
\frac{dg}{dt} &= -\frac{1}{\gamma}\frac{d\Omega}{d\gamma} = -\tfrac{3}{8}\,m^2e^2 \text{ ,, } 2c-2g,\\
\frac{dl}{dt} &= -2\frac{d\Omega}{da} + \tfrac{1}{2}\,e\frac{d\Omega}{de} + \tfrac{1}{2}\,\gamma\frac{d\Omega}{d\gamma} = \left(-\tfrac{15}{2} + \tfrac{3}{8} + \tfrac{3}{8}\right)m^2e^2\gamma^2 \text{ ,, } 2c-2g,
\end{aligned}$$

but this value of $\dfrac{dl}{dt}$ is, as will presently be seen, incomplete.

Writing $a+\delta a,\ e+\delta e$, &c., in place of $a,\ e$, &c., and observing that the divisor for the integration of the term in $2c-2g$ is $2(c-g),\ = -3m^2$, the first five equations give respectively

$$\begin{aligned}
\delta a &= 0,\\
\delta e &= -\tfrac{3}{8}\,\gamma^2 e \cos 2c-2g,\\
\delta\gamma &= +\tfrac{3}{8}\,\gamma e^2 \text{ ,, } 2c-2g,\\
\delta c &= +\tfrac{1}{8}\,\gamma^2 \sin 2c-2g,\\
\delta g &= +\tfrac{3}{8}\,e^2 \text{ ,, } 2c-2g.
\end{aligned}$$

The constant term in Ω is

$$= m^2a^2a''\left(\tfrac{1}{2} + \tfrac{3}{4}e^2 - \tfrac{3}{4}\gamma^2\right),$$

and this gives in

$$\frac{dl}{dt} = -2\frac{d\Omega}{da} + \tfrac{1}{2}e\frac{d\Omega}{de} + \tfrac{1}{2}\gamma\frac{d\Omega}{d\gamma},$$

a term

$$m^2\left(-1 - \tfrac{9}{4}e^2 + \tfrac{9}{4}\gamma^2\right) + \tfrac{3}{4}e^2 - \tfrac{3}{4}\gamma^2,$$

which is

$$= m^2\left(-1 - \tfrac{3}{2}e^2 + \tfrac{3}{2}\gamma^2\right).$$

Substituting for $e,\ \gamma$, their correct values $e+\delta e,\ \gamma+\delta\gamma$, it appears that $\dfrac{dl}{dt}$ contains the term

$$m^2\left(-\tfrac{3}{2}e\,\delta e + \tfrac{3}{2}\gamma\,\delta\gamma\right),$$

which is

$$= m^2\left(\tfrac{9}{16} + \tfrac{9}{16}\right)e^2\gamma^2 \cos 2c-2g = \tfrac{9}{8}\,m^2e^2\gamma^2 \text{ ,, } 2c-2g,$$

and joining to this the before-mentioned term

$$= -\tfrac{27}{8}\,m^2e^2\gamma^2 \text{ ,, } 2c-2g,$$

we find

$$\frac{dl}{dt} = \left(\tfrac{9}{8} - \tfrac{27}{8}\right)m^2e^2\gamma^2 \text{ ,, } 2c-2g,$$

whence, writing as above $l+\delta l$ for l, and integrating, we have

$$\delta l = -\tfrac{1}{16}\,e^2\gamma^2 \sin 2c-2g,$$

and it thus appears that the values of $\delta a,\ \delta e,\ \delta\gamma,\ \delta c,\ \delta g,\ \delta l$, agree with those obtained in my former Note.

467. EXPRESSIONS FOR PLANA'S e, γ IN TERMS OF THE ELLIPTIC e, γ.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXV. *(*1864–1865*), pp.* 265–271.*]*

The coefficient of $\sin cnt$ in Plana's expression for the true longitude v (see Plana, t. I. p. 574), putting therein $E' = e' = e'$, that is, neglecting the terms which depend on the variation of the solar excentricity, is

$$
\begin{aligned}
&= e\left(2 + \tfrac{3}{4}m^2 - \tfrac{11}{48}m^4 - \tfrac{2363}{1152}m^6 - \tfrac{19259}{4608}m^6 - \tfrac{511469}{36864}m^8\right)\\
&\quad + e^3\left(-\tfrac{1}{4} - 17m^2 - \tfrac{415}{32}m^4 - \tfrac{84539}{1152}m^6\right)\\
&\quad + e^5\left(\tfrac{1}{96} + \tfrac{1717}{96}m^2\right)\\
&\quad + e^7\left(\tfrac{6727}{1152}\right)\\
&\quad + e\gamma^2\left(-\tfrac{1}{4} - \tfrac{11}{16}m^2 + \tfrac{1487}{96}m^4 + \tfrac{7212}{144}m^6\right)\\
&\quad + e\gamma^4\left(\tfrac{11}{32} - \tfrac{8231}{384}m + \tfrac{4923}{72}m^2\right)\\
&\quad + e^3\gamma^2\left(-\tfrac{183}{16}\right)\\
&\quad + e\gamma^6\left(-\tfrac{1331}{144}\right)\\
&\quad + e\gamma^4\left(-\tfrac{3}{8} + \tfrac{135}{32}m + \tfrac{37}{12}m^2\right)\\
&\quad + e^3\gamma^4\left(\tfrac{4233}{144}\right)\\
&\quad + e\gamma^6\left(-\tfrac{157}{72}\right)\\
&\quad + ee'^2\left[\left(-\tfrac{3}{8} + \tfrac{3}{2} = \tfrac{9}{8}\right) - 9m^2 + \left(-\tfrac{1855}{128} + \tfrac{339}{32}\right) = \tfrac{3}{8}\,\tfrac{1}{32}m^4 + \left(-\tfrac{3045153}{1024} - \tfrac{267}{32}\right) = \tfrac{1}{32}\,\tfrac{3}{1024}m^6\right]\\
&\quad + e^3e'^2\left[\left(-\tfrac{4923}{96} - \tfrac{3}{8}\right) = -\tfrac{4131}{96}m^2\right]\\
&\quad + ee'^2\gamma^2\left[\left(\tfrac{8551}{192} - \tfrac{3309}{16}\right) = -\tfrac{15825}{96}m^2\right]\\
&\quad + ee'^4\left[\left(\tfrac{15}{8} - \tfrac{2925}{32}\right) = -\tfrac{1791}{32}m^4\right]\\
&\quad + ee^2\left(-\tfrac{4593}{1024}m^2\right)\\
&\quad + ee'^6\left(-\tfrac{25}{32}\right).
\end{aligned}
$$

Taking this to the fifth order only, and comparing it with the coefficient in the elliptic theory, we have

Plana.	Elliptic.
$e\left(2 - \tfrac{7}{12}m^2 - \tfrac{1027}{1152}m^4 - \tfrac{8235}{4608}m^6\right)$	$= e\,(2)$
$+e^3\left(-\tfrac{1}{4} - 17m^2\right)$	$+e^3\left(-\tfrac{1}{4}\right)$
$+e^5\left(\tfrac{1}{96}\right)$	$+e^5\left(\tfrac{1}{96}\right)$
$+e\gamma^2\left(-\tfrac{1}{4} - \tfrac{23}{12}m^2\right)$	
$+e\gamma^4\left(-\tfrac{3}{8}\right)$	
$+e^3\gamma^2\left(\tfrac{11}{16}\right)$	
$+ee'^2\left(-9m^2\right).$	

The coefficient of $\sin gnt$ in Plana's expression for the latitude (see t. I. p. 704) is

$$
\begin{aligned}
&= \gamma\left(1 + \tfrac{15}{16}m^2 + \tfrac{121}{48}m^4 - \tfrac{1023}{1152}m^6\right)\\
&\quad + \gamma e^2\left(-\tfrac{1}{2} - \tfrac{15}{32}m^2 - \tfrac{257}{96}m^4\right)\\
&\quad + \gamma e^4\left(\tfrac{5}{16} + \tfrac{339}{96}m^2\right)\\
&\quad + \gamma e^6\left(\tfrac{3}{16}\right)\\
&\quad + \gamma^3\left(-\tfrac{1}{8} - \tfrac{31}{96}m^2 + \tfrac{1119}{96}m^4\right)\\
&\quad + \gamma^3 e^2\left(\tfrac{4}{16} - \tfrac{4133}{96}m^2\right)\\
&\quad + \gamma^5\left(\tfrac{3}{32}\right)\\
&\quad + \gamma e'^2\left(\tfrac{27}{32}m^2 - m^4\right).
\end{aligned}
$$

But according to the calculation of Prof. Adams (quoted by M. Delaunay, *Comptes Rendus*, t. LIV. (1862), this should be

$$\begin{aligned}
&= \gamma\big(1 + \tfrac{13}{16}m^2 - \tfrac{1}{12}m^4 - \tfrac{6135}{1152}m^6 - \tfrac{4351477}{2304}m^6\big)\\
&\quad + \gamma e^2\big(-\tfrac{1}{2} - \tfrac{15}{32}m^2 - \tfrac{249}{96}m^4\big)\\
&\quad + \gamma e^4\big(\tfrac{5}{16} + \tfrac{15}{96}m^2\big)\\
&\quad + \gamma e^6\big(\tfrac{3}{16}\big)\\
&\quad + \gamma^3\big(-\tfrac{1}{8} - \tfrac{2}{3} + \tfrac{1119}{96}m^4\big)\\
&\quad + \gamma^3 e^2\big(\tfrac{4}{16} + \tfrac{4133}{96}m^2\big)\\
&\quad + \gamma^5\big(\tfrac{3}{32}\big)\\
&\quad + \gamma e'^2\big(\tfrac{27}{32}m^2 - \tfrac{1}{16}m^4\big)\,.
\end{aligned}$$

Adopting this as the true expression according to Plana's theory, taking it to the fifth order only, and comparing with the elliptic value of the same coefficient, we have

Plana.	Elliptic.
$\gamma\ \big(1 + \tfrac{13}{16}m^2 - \tfrac{1}{12}m^4\big)$	$= \gamma\ (1)$
$+\gamma e^2\ \big(-\tfrac{1}{2} - \tfrac{15}{32}m^2\big)$	$+\gamma e^2\ \big(-\tfrac{1}{2}\big)$
$+\gamma e^4\ \big(\tfrac{5}{16}\big)$	$+\gamma e^4\ \big(\tfrac{5}{16}\big)$
$+\gamma^3\ \big(-\tfrac{1}{8} - \tfrac{2}{3}\big)$	$+\gamma^3\ \big(-\tfrac{1}{8}\big)$
$+\gamma^3 e^2\ \big(\tfrac{4}{16}\big)$	$+\gamma^3 e^2\ \big(\tfrac{4}{16}\big)$
$+\gamma^5\ \big(\tfrac{3}{32}\big)$	$+\gamma^5\ \big(\tfrac{3}{32}\big)$
$+\gamma e'^2\ \big(\tfrac{27}{32}m^2\big)$	.

We have thus two equations for the determination of Plana's e, γ in terms of the elliptic e, γ. And the solution of these equations give

$$\begin{aligned}
&\qquad\quad \text{Elliptic.}\\
e\,(\text{Plana}) = e\ &\big(1 - \tfrac{3}{8}\,m^2 + \tfrac{27}{32}m^4 + \tfrac{521}{1152}m^6\big)\\
&+e^3\ \big(\tfrac{17}{16}m^2\big)\\
&+\gamma^2 e\ \big(\tfrac{1}{4} + \tfrac{23}{12}m^2\big)\\
&+\gamma^2 e^3\ \big(-\tfrac{1}{2}\big)\\
&+\gamma e\ \big(\tfrac{1}{4}\big)\\
&+ee'^2\ \big(\tfrac{9}{2}m^2\big)\,,\\
\gamma\,(\text{Plana}) = \gamma\ &\big(1 - \tfrac{3}{32}m^2 + \tfrac{2}{3}m^4\big)\\
&+\gamma e^2\ \big(-\tfrac{3}{16}\big)\\
&+\gamma e^4\ \big(-\tfrac{1}{16}m^2\big)\\
&+\gamma e^6\ \big(\tfrac{11}{16}\big)\\
&+\gamma^3\ \big(\tfrac{2}{3}\big)\\
&+\gamma^5 e^2\ \big(-\tfrac{1}{2}m^2\big)\,.
\end{aligned}$$

I annex the verification of these expressions; we have

Plana.	Elliptic.
$e\left(2+\frac{3}{4}m^2-\frac{11}{48}m^4-\frac{2363}{1152}m^6\right)$	$=e\left(2-\frac{3}{8}m^2+\frac{27}{32}m^4+\frac{521}{1152}m^6\right.$
	$+\frac{4}{3}m^4-\frac{8}{3}m^6$
	$\left.-\frac{19}{12}m^6-\frac{6135}{1152}m^6\right)$
$+e^3\left(\frac{17}{16}m^2\right)$	$=e^3\left(\frac{17}{16}m^2\right)$
$+e\gamma^2\left(\frac{1}{4}+\frac{23}{12}m^2\right)$	$=e\gamma^2\left(\frac{1}{4}+\frac{23}{12}m^2\right)$
	$+\frac{4}{3}m^4$
$+e\gamma^4\left(-\frac{3}{8}\right)$	$=e\gamma^4\left(-\frac{3}{8}\right)$
$+e\gamma^6\left(\frac{11}{16}\right)$	$=e\gamma^6\left(\frac{11}{16}\right)$
$+ee'^2\left(-9m^2\right)$	$=ee'^2\left(-9m^2\right),$
$e^3\left(-\frac{1}{4}-17m^2\right)$	$=e^3\left(-\frac{1}{4}+\frac{17}{16}m^2\right)$
	$-17m^2$
$e^5\left(\frac{1}{96}\right)$	$=e^5\left(\frac{1}{96}\right)$
$e^3\gamma^2\left(-\frac{1}{8}-\frac{23}{16}m^2\right)$	$=e^3\gamma^2\left(-\frac{1}{8}-\frac{23}{16}m^2\right)$
	$+\frac{4}{3}m^4$
$e^3\gamma^4\left(\frac{11}{16}\right)$	$=e^3\gamma^4\left(\frac{11}{16}\right)$
$e^5\gamma^2\left(-\frac{1}{8}\right)$	$=e^5\gamma^2\left(-\frac{1}{8}\right)$
$ee'^2\left(-9m^2\right)$	$=ee'^2\left(-9m^2\right),$

whence, adding, we have the first equation.

And, moreover,

Plana.	Elliptic.
$\gamma\left(1+\frac{13}{16}m^2-\frac{1}{12}m^4\right)$	$=\gamma\left(1-\frac{3}{32}m^2+\frac{2}{3}m^4\right.$
	$+\frac{4}{3}m^4-\frac{2}{3}m^4$
	$+\gamma e^2\left(-\frac{23}{12}m^2\right)$
	$+\gamma e^2\left(\frac{2}{3}\right)$
	$+\gamma e^4\left(-\frac{1}{2}m^2\right)$
	$+\gamma^3\left(-\frac{2}{3}\right),$
$\gamma e^2\left(\frac{3}{32}\right)$	$=\gamma e^2\left(\frac{3}{32}\right)$
$\gamma\left(-\frac{1}{2}+\frac{15}{32}m^2\right)$	$=\gamma\left(-\frac{1}{2}+\frac{15}{32}m^2\right)$
$\gamma^3e^2\left(\frac{4}{16}\right)$	$=\gamma^3e^2\left(\frac{4}{16}\right)$
$\gamma^5\left(\frac{3}{32}\right)$	$=\gamma^5\left(\frac{3}{32}\right)$
$\gamma e^2\left(\frac{27}{32}m^2\right)$	$=\gamma e^2\left(\frac{27}{32}m^2\right),$

whence, adding, we have the second equation.

It may be noticed that, taking the foregoing expressions only as far as the third order, we have

$$\begin{array}{lcl} \text{Plana.} & & \text{Elliptic.} \\ e & = & e\left(1+\frac{1}{4}\gamma^2-\frac{3}{8}m^2\right), \\ \gamma & = & \gamma. \end{array}$$

And moreover that, attending only to the terms which are independent of m, we have

$$e=e\left(1+\tfrac{1}{4}\gamma^2-\tfrac{3}{8}e^2\gamma^2+\tfrac{1}{4}\gamma^4\right),$$
$$\gamma=\gamma\left(1-\tfrac{3}{16}e^2+\tfrac{4}{16}e^2\gamma^2-\tfrac{1}{2}\gamma^4\right).$$

which are formulae that may be found useful.

468. Addition to Second Note on the Lunar Theory.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXVII. *(*1866–1867*), pp.* 267–269.*]*

Writing as in my Second Note, *Monthly Notices*, Vol. XXV., pp. 203–207 (May 1865), [466], for the Moon,

a, the mean distance,
e, the excentricity,
γ, the tangent of the inclination,
l, the mean longitude,
c, the mean anomaly,
g, the mean distance from node,

I obtained by the ordinary method of the variation of the elements, from the constant term of R and the term involving $\cos(2c-2g)$, the following expressions of the variations,

$$\begin{aligned}
\delta a &= 0,\\
\delta e &= -\tfrac{3}{8}\gamma^2 e \cos 2c-2g,\\
\delta\gamma &= +\tfrac{3}{8}\gamma e^2 \;\;,,\;\; 2c-2g,\\
\delta c &= +\tfrac{1}{8}\gamma^2 \sin 2c-2g,\\
\delta g &= +\tfrac{3}{8}e^2 \;\;,,\;\; 2c-2g,\\
\delta l &= +\tfrac{1}{16}\gamma^2 e^2 \;\;,,\;\; 2c-2g,
\end{aligned}$$

viz. if in the elliptic expressions of the radius vector, longitude, and latitude, we apply to a, e, γ, c, g, l, the foregoing increments, we obtain to the fourth order in (e,γ) the portions independent of m in the expressions of the radius vector, latitude, and longitude. I wish to notice that the results, to the very limited extent to which they go, agree with those obtained by M. Delaunay in his "Théorie du Mouvement de la Lune," from his 49th operation, the object of which is to take away the term (63) of R, that is the term involving $\cos(2c-2g)$. The formulae (see vol. I. p. 788), taken only to the necessary degree of approximation are

$$\begin{array}{rcl}
a & \text{replaced by} & a,\\
e^2 & ,, & e^2-\tfrac{3}{4}\gamma e^2\cos 2g,\\
\gamma^2 & ,, & \gamma^2+\tfrac{3}{4}\gamma^2 \;\;,,\;\; 2g,\\
l & ,, & l-\tfrac{1}{8}\gamma^2\sin 2g,\\
h+g+l & ,, & h+g+l+\tfrac{3}{8}\gamma^2 e^2 \;\;,,\;\; 2g,\\
h & ,, & h+\tfrac{3}{8}e^2 \;\;,,\;\; 2g,
\end{array}$$

which, observing that

$$\begin{aligned}
\gamma\,(\text{Del.}) &= \tfrac{1}{2}\gamma \text{ (for present purpose)},\\
l &= c,\\
g+h &= g,\\
h+g+l &= l,
\end{aligned}$$

and therefore

$$g \;=\; -(c-g),$$

become

$$\begin{array}{rcl}
a & \text{replaced by} & a,\\
e^2 & ,, & e^2-\tfrac{3}{8}\gamma^2 e^2\cos 2c-2g,
\end{array}$$

$$\begin{array}{rll}\gamma^2 & ,, & \gamma^2 + \tfrac{3}{4}\gamma^2 e^2 \;,,\; 2c-2g,\\ c & ,, & c + \tfrac{1}{8}\gamma^2 \sin 2c-2g,\\ l & ,, & l - \tfrac{1}{16}\gamma^2 e^2 \;,,\; 2c-2g,\\ l-g & ,, & l-g-\tfrac{3}{8}e^2 \;,,\; 2c-2g,\end{array}$$

the last of which may be changed into

$$g \quad ,, \quad g + \tfrac{3}{8}e^2 \;,,\; 2c-2g,$$

or if the new values of a, e, γ, c, g, l, are called $a+\delta a$, $e+\delta e$, $\gamma+\delta\gamma$, $c+\delta c$, $g+\delta g$, $l+\delta l$, then the increments δa, δe, $\delta\gamma$, δc, δg, δl, have the values given above. The process of my Second Note, taken as a first transformation, has in fact the object of removing the term $\cos(2c-2g)$, and to the degree of approximation regarded, the result is not affected by the previous transformations, or by the substitution, t. II. p. 800, introducing for a, e, γ, their standard elliptic values.

469. On an Expression for the Angular Distance of Two Planets.

[From the Monthly Notices of the Royal Astronomical Society, vol. XXVII. *(*1866–1867*), pp.* 312–315.*]*

If for the planet m, referred to any fixed plane and origin of longitudes, we have

$$\begin{array}{l} v,\ \text{the longitude in orbit,}\\ \theta,\ \text{the longitude of node,}\\ \phi,\ \text{the inclination,}\end{array}$$

and similarly for the planet m' referred to the same fixed plane and origin of longitudes, if the corresponding quantities are v', θ', ϕ'; then the angular distance of the two planets will of course be expressible in terms of v, θ, ϕ, v', θ', ϕ', but I am not aware that the actual expression has been given. To obtain it in the most simple manner, I write further for the planet m:

$$\begin{array}{l} \theta+x,\ \text{the reduced longitude,}\\ y,\ \text{the latitude,}\\ z,\ \text{the distance from node,}\end{array}$$

so that z $(=v-\theta)$, x, y, are the hypothenuse, base, and perpendicular of a right-angled spherical triangle, the base angle of which is $=\phi$. And similarly $\theta'+x'$, y', z', have the like significations for the planet m'. I write also r, r', for the distances of the two planets respectively.

This being so, the rectangular coordinates of the planet m are

$$\begin{array}{l} r\cos y\,\cos(\theta+x),\\ r\cos y\,\sin(\theta+x),\\ r\sin y.\end{array}$$

But observing that from the right-angled triangle we have

$$\begin{aligned}\cos z &= \cos x\,\cos y,\\ \cos\phi &= \tan x\,\cot z,\\ \sin x &= \cot\phi\,\tan y,\\ \sin y &= \sin\phi\,\sin z,\end{aligned}$$

and therefore also

$$\sin x\cos y \;=\; \cot\phi\,\sin y, \;=\; \cos\phi\,\sin z,$$

the expressions for the coordinates become

$$\begin{aligned}&r\,(\cos z\,\cos\theta-\sin\phi\,\sin\theta\,\cos\phi),\\&r\,(\cos z\,\sin\theta+\sin z\,\cos\theta\,\cos\phi),\\&r\,(\sin z\,\sin\phi).\end{aligned}$$

Forming the analogous expressions for the coordinates of m', then if H be the angular distance of the two planets, we deduce at once the expression for $\cos H$, viz. this is

$$\begin{aligned}rr'\cos H=&(\cos z\,\cos\theta-\sin z\,\sin\theta\,\cos\phi)\,(\cos z'\,\cos\theta'-\sin z'\,\sin\theta'\,\cos\phi')\\&+(\cos z\,\sin\theta+\sin z\,\cos\theta\,\cos\phi)\,(\cos z'\,\sin\theta'+\sin z'\,\cos\theta'\,\cos\phi')\\&+(\sin z\,\sin\phi)\,(\sin z'\,\sin\phi'),\end{aligned}$$

or, multiplying out, this is

$$\begin{aligned}rr'\cos H=&\cos z\,\cos z'\,\cos(\theta-\theta')\\&+\cos z\,\sin z'\,\sin(\theta-\theta')\,\cos\phi'\\&-\sin z\,\cos z'\,\sin(\theta-\theta')\,\cos\phi\\&+\sin z\,\sin z'\,(\cos(\theta-\theta')\,\cos\phi\,\cos\phi'+\sin\phi\,\sin\phi'),\end{aligned}$$

say this is

$$rr'\cos H\ =\ A\,\cos z\,\cos z'+B\,\cos z\,\sin z'+C\,\sin z\,\cos z'+D\,\sin z\,\sin z',$$

viz. it is

$$\begin{aligned}&\cos(z-z')\cdot\tfrac{1}{2}A+\tfrac{1}{2}D\\&\quad+\sin(z-z')\cdot\tfrac{1}{2}B-\tfrac{1}{2}C\\&\quad+\cos(z+z')\cdot\tfrac{1}{2}A-\tfrac{1}{2}D\qquad z+z'=v+v'-\theta-\theta',\\&\quad+\sin(z+z')\cdot\tfrac{1}{2}B+\tfrac{1}{2}C.\end{aligned}$$

But we have

$$z-z'\ =\ v-v'-\theta+\theta',$$

[Paper 469 continues on subsequent pages, beyond the present chunk.]